— Chaos —

→ Predrag on the standard model:
- QCD is highly nonlinear → more interesting
- QED is 'less' difficult, mostly linear
- Yang-Mills for standard model is generalization of electrodynamics

→ Chaos: "you touch something, it falls apart"

→ It is more like statistical mechanics than classical Mechanics

→ you have to consider all admissible cases of all possible states, and do probability with it

→ Even when you have high dimensional systems (e.g., a membrane) with complex interactions (like atomic bending) will admit smooth, averaged solutions via Fourier / Functional analysis

⇒ a good way to view NS: replace the 'imaginary' mass particles / mass elements from Eulerian Formulation, write the state of the system as a fourier series

→ convert PDEs into an 'infinite tower' of ODEs organized by Fourier mode

→ Consider Meteorological Systems

# Dims:	∞	→	11	→	3
desc.	Full behavior		N-S EQNs		Symmetry simplification

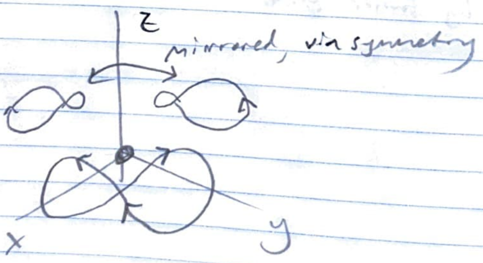
⇒ the Lorenz System, to model a simplified atmospheric dynamical system, reads

$r > 1$

$$\dot{x} = \sigma(y - x)$$
$$\dot{y} = rx - y - xz$$
$$\dot{z} = xy - bz$$

→ 3D+: It now becomes much less likely that there is some topological structure that enforces a trajectory returning to the same point, that is, orbits in phase space are highly special/rare cases → Markov process "is allowed to forget the past", finite time

→ Symmetry: $(x, y) \to (-x, -y)$



→ use symmetries of the equations to elucidate properties of the system

→ calculate equilibria (x^*, y^*, z^*)

factor out simple root, use quadratic formula on factored cubic

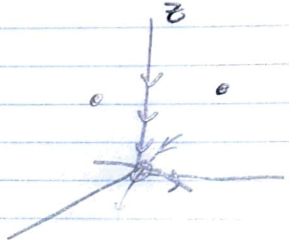
$(0,0,0)$; $(+\sqrt{b(r-1)}, +\sqrt{b(r-1)}, r)$;

$(-\sqrt{b(r-1)}, -\sqrt{b(r-1)}, r)$

→ Stability matrix:

$$A_{ij} = \begin{pmatrix} -\sigma & \sigma & 0 \\ r-z & -1 & -x \\ y & x & -b \end{pmatrix}$$

$$A(0,0) = \begin{pmatrix} -\sigma & \sigma & 0 \\ r & -1 & 0 \\ 0 & 0 & -b \end{pmatrix} \Rightarrow$$

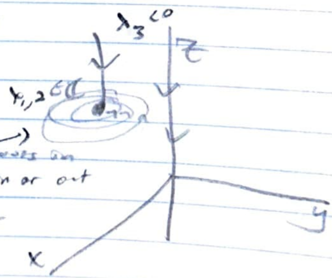


z-axis: → invariant subspace under rotations

$r<1$: diffusive convection, viscous system, fully stable

$r=1$: bifurcation

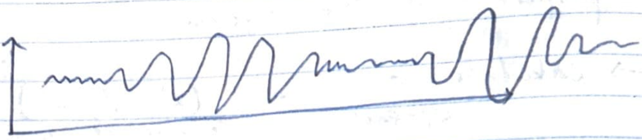
$r>1$: instability forms



→ fall in along λ_3, spiral out along $\lambda_{1,2}$ plane

→ eventually, cross invariant z-axis, causes jump

→ consider the motion in only z:



↳ deterministic eqn, but
a-periodic solution!